DraftFM: Implementation and Reproducibility Reference

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Abstract

This is the companion reference for "DraftFM: Zero-Shot Drafting of Unseen *Magic: The Gathering* Sets from Public Card Features." The main paper is the scientific story: modeling decisions, the pre-registered protocol, results, and an honest self-assessment of what it does and doesn't show. This document is the short, structured "how it's actually built and reproduced" reference instead – released code layout, the provenance mechanics behind every number in the main paper, and the frozen identifiers that pin the evaluation. Read the main paper first.

1 Release contents and status

The repository contains the training and evaluation code, the frozen protocol and metric implementations, the experiment ledger, selected summary artifacts, and generated paper outputs. The publication archive is intended to add the exact model weights and cached per-pick prediction files used by the paper. No archive DOI is assigned yet, so this document does not claim those large artifacts are already public. Raw 17Lands bytes for the MSH test snapshot will be considered for deposit only after a courtesy note to 17Lands; other inputs are re-derivable from their public sources and manifests.

2 Provenance mechanics

Training runs append records to an experiment ledger whose fields vary by run family; the DraftFM records include the run id, git sha, seed, framework version, config, counters, timing, and artifact hash. Frozen evaluation summaries are stored separately rather than appended to that ledger. A release manifest pins the exact run ids consumed by the table generator, and generated result cells carry source comments where a single run supplies the value. Primary table and headline numbers read from generated macros; secondary analysis numbers are produced by the cited scripts and cached machine-readable outputs.

Training's MPS backend is not trusted by default: every launch runs a CPU-vs-accelerator parity check on losses and gradient norms before training proceeds, LayerNorm is composed from primitive ops rather than the fused kernel (whose backward pass is wrong on some chip/OS combinations), and a watchdog skips non-finite gradient steps during training and halts on recurrence rather than silently continuing.

- `experiments/ledger.jsonl` – the append-only run registry
- `scripts/make_paper_tables.py` – emits every table and the scaling-figure data from ledger + run records; regenerate after any new run lands

- `figures/make_verdict_panels.py` – computes the real, live-scored numbers in the illustrative verdict-panel figures via the same `mtga.models.registry` / `mtga.draft_api.HUB` code path the deployed overlay uses
- `figures/fetch_example_cards.py` – fetches the card images used in those figures directly from Scryfall's hosted URLs, writing `figures/cards/manifest.json` with the exact source URL, artist, and copyright line per card

3 Frozen identifiers

Protocol tag	`eval-protocol-v1.1`
Data-manifest content hash	`443bd25f72ee`
Featurizer-manifest content hash	`e1313cadd03b`
F-dev run id	`20260704_135822_f_dev`
F-full run id	`20260705_110743_f_full`

The MSH snapshot was fetched at 2026-07-07 02:30:04 EDT. Its identifiers are:

```
sha256:  013df16b8994534f69ed63c87ab684ac
         afc5f4cbe82982264b0fc111dbb2183a
ETag:  "252007adfbca6f026766527823ffa6d5-8"
```

The frozen pass recorded 1,607,786 rows, 6,286 high-win-rate-slice drafts, a modern schema, complete first picks, and a 100% pack-name join rate; every declared gate passed.